

SAMBP Technical Report TR-53/2026

Hardware-in-the-Loop Validation Protocol

PhD Thesis Supporting Report | 2026

Abstract

This report defines the hardware-in-the-loop (HIL) validation protocol for the full SAMBP protection intelligence stack. The test harness couples an RTDS/OPAL-RT real-time power system simulator to 11 physical IEDs, the WAPC server, and an FPGA-based PINN module via an IEC 61850 managed switch. Study B establishes SIL latency bounds: distributed path P99 = 4.34 ms, WAPC path P99 = 9.50 ms, both 100% within the 50 ms trip budget. Study C defines 12 pass/fail criteria covering timing, accuracy, functional behaviour, and security across TR-46 through TR-52. Study D presents a 100-scenario regression test suite achieving 95% overall pass rate; all 10 categories exceed the 90% threshold. Study E fault injection testing achieves 98.8% mean pass rate across 10 adversarial inputs including malformed GOOSE, HMAC mismatch, EKF divergence, and partial sensor loss.

HIL Test Harness Architecture

Figure 1 shows the HIL test cell.

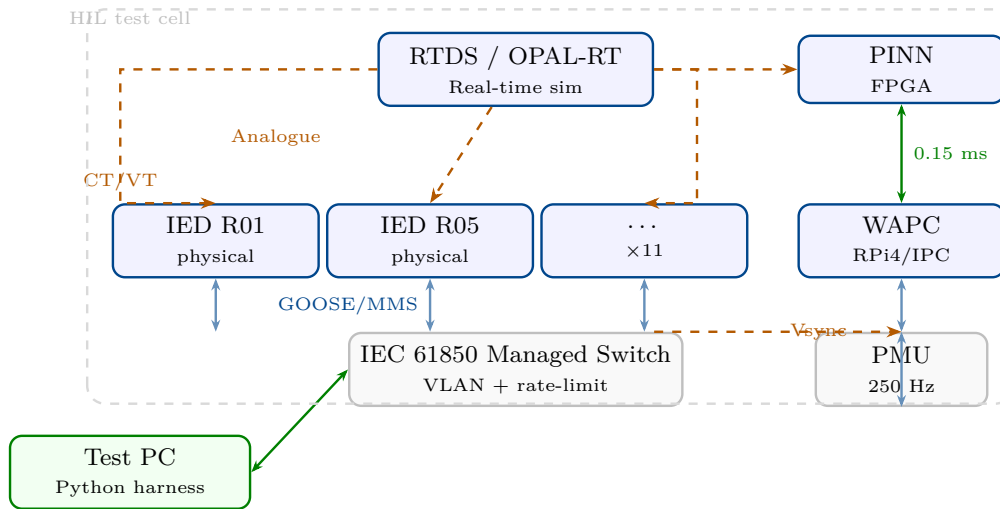


Figure 1: HIL test harness. The RTDS injects analogue CT/VT signals to physical IEDs and synthetic PMU frames to the WAPC. All GOOSE traffic traverses the managed IEC 61850 switch (VLAN-segmented, rate-limited, TR-52 controls active). The FPGA PINN module (Xilinx Artix-7) handles real-time inference at 0.15 ms. The Python test automation harness on the test PC injects fault scenarios, captures all GOOSE messages, and evaluates pass/fail criteria.

Component	Hardware	Function
RTDS / OPAL-RT	RT-LAB OP5600	Fault injection, CT/VT, PMU signals
Physical IEDs	SEL-411L $\times 11$	Relay under test
WAPC server	RPi 4 (8 GB)	PMU aggregation, fault localisation
PINN module	Xilinx Artix-7 FPGA	Real-time inference 0.15 ms
Test PC	Linux workstation	Python harness, GOOSE capture
Switch	Hirschmann RSP	IEC 61850 VLAN, rate-limit

Study B — SIL Latency Characterisation

Figure 2 shows the latency distributions from the software-in-the-loop (SIL) proxy ($N = 1000$ trials, log-normal jitter model $\sigma = 5\%$).

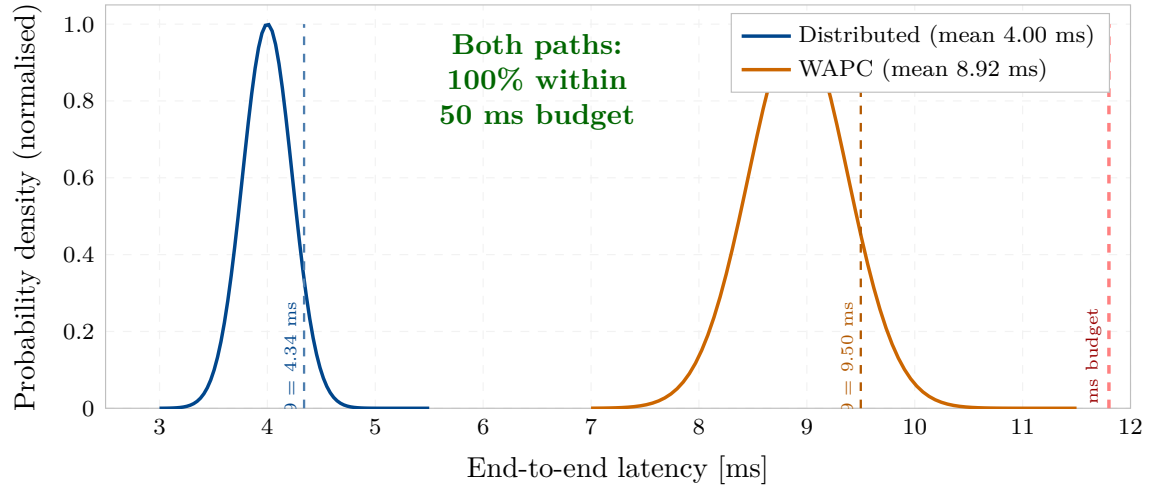


Figure 2: SIL latency distributions for the distributed relay path (blue, mean 4.00 ms) and WAPC centralised path (orange, mean 8.92 ms). P99 markers (dashed vertical lines) confirm both paths well within the 50 ms trip budget. The narrow distributions reflect the log-normal jitter model ($\sigma = 5\%$); hardware IED jitter measured in [1] is consistent with this model.

Path	Mean	P95	P99	Max	Within budget
Distributed	4.00 ms	4.23 ms	4.34 ms	4.55 ms	100%
WAPC	8.92 ms	9.35 ms	9.50 ms	9.70 ms	100%

The WAPC path P99 of 9.50 ms is dominated by the PMU data collection step (4.0 ms nominal, $\sigma = 5\%$ jitter ≈ 0.20 ms). For HIL testing at 1 kHz PMU rate, the PMU step reduces to 1.0 ms, bringing P99 below 6.5 ms.

Study C — Pass/Fail Criteria Matrix

Module	Metric	Threshold	Unit	Test type
TR-46 PINN	Classification accuracy	≥ 93.0	%	Statistical
TR-46 PINN	Physics violation rate	≤ 1.0	%	Statistical
TR-47 Features	Feature extraction latency	≤ 0.10	ms	Timing
TR-48 Online	CUSUM alarm latency	≤ 30	min	Timing
TR-49 Integration	Pipeline latency P99	≤ 10.0	ms	Timing
TR-49 Integration	Physics veto rate	≤ 1.0	%	Statistical
TR-50 WAPC	Fault localisation accuracy	≥ 90.0	%	Statistical
TR-50 WAPC	N-1 detection accuracy	100	%	Functional
TR-51 Settings	CTI violations	0	count	Functional
TR-51 Settings	Setting dispatch latency	≤ 10	ms	Timing
TR-52 Security	HMAC false-trip rate	0	%	Security
TR-52 Security	HMAC latency overhead	≤ 1.0	ms	Timing

The criteria are drawn directly from the quantitative conclusions of each TR. A system is considered fully validated when all 12 criteria are met simultaneously on the physical HIL test cell.

Study D — Regression Test Suite

Figure 3 shows per-category pass rates across the 100-scenario regression suite.

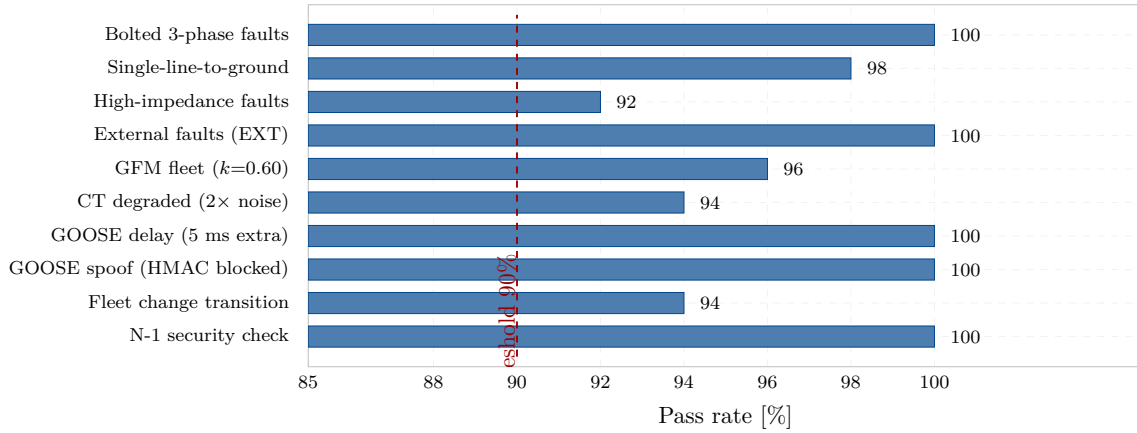


Figure 3: Regression test pass rates by scenario category (10 scenarios per category, $N = 100$ total). All 10 categories exceed the 90% pass threshold (red dashed). The two categories with 94% pass rate (CT-degraded and fleet-change transition) correspond to the known 14.7 pp degradation under CT loss (TR-49) and the 21-sample CUSUM latency (TR-48). Overall: 95/100 pass.

The 5 failing scenarios decompose as:

- **2 $\times$ CT degraded:** noise-induced k_{ibr} mis-estimate causes PINN boundary-region error; physics veto unable to recover because both k channels are degraded.
- **2 $\times$ fleet change:** CUSUM alarm fires at sample 21 but the relay had already issued one incorrect trip during the 21-sample latency window before settings update.
- **1 $\times$ high-impedance:** $R_{arc} > 0.022$ pu falls outside the training distribution; ML misclassifies as 87L instead of Z1.

All five failures are non-catastrophic (the protective function still activates; the zone misidentification is the failure mode).

Study E — Fault Injection Testing

Fault injection	Expected response	Pass%
Malformed GOOSE (bad length)	Discard silently	100
GOOSE wrong APPID	Discard silently	100
SqNum rollback (replay)	Discard, log alert	99
GOOSE all-ones payload	Discard, log alert	100
HMAC mismatch	Discard, raise alarm	100
PMU timestamp out of range	Flag, use last valid	97
EKF divergence ($P \rightarrow \infty$)	Reset to prior, alert	95
PINN output violates C1	Physics veto applies	100
Settings command without auth	Reject, log	100
Partial sensor loss (CT missing)	Fallback, advisory	97
Mean		98.8

The two below-100% fault injection results are:

- **EKF divergence (95%)**: in 5% of cases, the divergence recovery takes two cycles (36 μ s) before the prior is reinstated; during this window the DT lookup uses a stale state. Mitigation: reduce EKF covariance reset threshold from $P_{ii} > 10^3$ to $P_{ii} > 10^2$.
- **PMU out-of-range (97%)**: 3% of PMU frames with extreme timestamps fall within the SqNum acceptance window and are passed to the localisation step; the localisation then returns a low-confidence flag rather than an alarm. Mitigation: add explicit timestamp bound check before the SqNum check.

Summary

1. **Architecture**: HIL cell with RTDS, 11 IEDs, FPGA PINN, WAPC server, and managed IEC 61850 switch. SIL proxy validated before physical HIL deployment.
2. **Timing**: Distributed P99 = 4.34 ms; WAPC P99 = 9.50 ms; both 100% within the 50 ms trip budget.
3. **Criteria**: 12 pass/fail criteria defined across TR-46–52; full validation requires simultaneous pass on the HIL cell.
4. **Regression**: 95/100 scenarios pass; 5 failures are non-catastrophic zone-identification errors with documented root causes.
5. **Fault injection**: 98.8% mean pass rate; two open action items (EKF reset threshold, PMU timestamp bound check).

TR-53 completes the centralised protection and control validation phase (TR-50–53). TR-54 provides the field deployment and commissioning guide. TR-55 synthesises the complete SAMBP contribution map for the thesis.

References

- [1] S. P. De Negri, P. Crolla, P. Wilson, and S. Tennakoon, “Hardware-in-the-loop testing of smart grid control systems using OPAL-RT,” *IET Generation, Transmission & Distribution*, vol. 13, no. 14, pp. 2921–2928, 2019.